AgentHire: A Multi-Agent Based Intelligent Framework for Comprehensive Interview Preparation

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Abstract*— The contemporary landscape of engineering education exhibits a significant disjunction between academic preparation and industry requirements, particularly in the domain of technical interviews. This research paper introduces AgentHire, an innovative multi-agent artificial intelligence system designed to bridge this critical gap through a sophisticated theoretical framework. The system employs a coordinated ecosystem of five specialized intelligent agents that collaboratively simulate the complete interview preparation lifecycle. Each agent embodies a distinct cognitive specialization, working in concert to provide personalized, adaptive, and comprehensive preparation experiences. The theoretical underpinnings of AgentHire draw from educational psychology, cognitive science, and artificial intelligence research, creating a novel paradigm for career readiness training. The system's architecture facilitates dynamic job market analysis, personalized content generation, realistic interview simulation, and iterative feedback mechanisms, all grounded in established pedagogical principles. This paper presents a comprehensive theoretical exploration of the system's design philosophy, implementation methodology, and potential educational impact, positioning AgentHire as a transformative approach to engineering career development in the digital age.

Keywords — Multi-Agent Artificial Intelligence, Interview Preparation Systems, Educational Technology, Engineering Education, Personalized Learning, Cognitive Apprenticeship, Career Readiness, Intelligent Tutoring Systems

I. Introduction

The transition from academic environments to professional workplaces represents one of the most challenging phases in an engineering student's educational journey. Current empirical evidence suggests that approximately seventy-eight percent of engineering graduates experience significant anxiety when facing technical interviews, despite possessing adequate theoretical knowledge and technical capabilities. This phenomenon reveals a critical gap in contemporary engineering education: the disconnect between academic learning and professional assessment methodologies. Traditional interview preparation approaches, including standardized test preparation, peer-based mock interviews, and career center workshops, demonstrate substantial limitations in addressing the complex, multi-dimensional nature of modern technical interviews.

The fundamental challenge resides in the dynamic and personalized nature of technical interviews, which assess not only theoretical knowledge but also problem-solving methodologies, communication skills, cultural alignment, and psychological resilience. Existing preparation methods typically operate within standardized, one-size-fits-all paradigms that fail to account for individual learning styles, varying target roles, company-specific requirements, and evolving industry trends. This standardization creates preparation experiences that are often misaligned with actual interview scenarios, leading to inadequate readiness and suboptimal performance outcomes.

AgentHire emerges as a theoretical solution to these systemic challenges through its innovative multi-agent artificial intelligence architecture. The system

conceptualizes interview preparation as a complex adaptive process requiring multiple specialized cognitive functions working in coordination. Rather than employing a monolithic artificial intelligence approach, AgentHire distributes cognitive responsibilities across five distinct intelligent agents, each specializing in a critical aspect of interview preparation. This distributed cognitive model mirrors the collaborative nature of professional engineering environments while providing comprehensive, personalized preparation experiences.

The theoretical foundation of AgentHire draws from several established educational paradigms, including cognitive apprenticeship theory, situated learning models, and adaptive learning frameworks. By integrating these educational principles with advanced artificial intelligence capabilities, the system creates a preparation environment that closely simulates real-world interview scenarios while providing structured learning pathways and personalized feedback mechanisms. This integration represents a significant advancement in educational technology, particularly in the domain of professional skill development

This research paper presents a comprehensive theoretical exploration of the AgentHire system, examining its architectural philosophy, agent specifications, implementation methodology, and potential educational impact. The paper contributes to the growing body of research on artificial intelligence in education while proposing a novel framework for addressing the critical challenge of interview preparation in engineering education.

II. Literature Survey

Significant research has been conducted on applying artificial intelligence and machine learning techniques to interview preparation and career readiness systems [1, 2, 3]. Many studies in this domain focus on the use of Large Language Models (LLMs) and conversational AI for simulating interview scenarios. For instance, some approaches have utilized dialog systems with reflective learning capabilities [1], interview dialog generators trained on real transcripts [2], and multi-agent conversation frameworks [3]. Techniques such as sentiment analysis, emotional recognition, and behavioral assessment have been widely applied in these systems [6, 8, 9].

Multi-agent AI systems have emerged as a powerful paradigm for complex educational applications, often reducing the need for monolithic architecture designs [3, 4]. Researchers have explored various multi-agent frameworks, including AutoGen for conversation-based systems [3], AutoAgents for dynamic agent generation [4], and specialized domain-specific architectures. Some studies have applied transfer learning by adapting general multi-agent frameworks to specific domains like education and career preparation [3, 4]. While multi-agent systems often significantly improve task specialization and system robustness, their performance can be influenced by factors such as agent coordination efficiency, communication protocols, and domain adaptation challenges [3, 4].

Recent advancements in immersive technologies have expanded the capabilities of interview preparation systems. Photorealistic AI-powered Metahuman avatars in Virtual and Augmented Reality environments have been developed to enhance user engagement and psychological preparedness [7]. These systems measure psychological factors including interview anxiety, social presence, self-esteem, and intrinsic motivation during mock interviews [7]. However, studies have shown that the technological medium must be carefully balanced with pedagogical effectiveness, as highly realistic avatars might not necessarily translate to better learning outcomes compared to well-structured content and feedback systems [7].

From a technical implementation perspective, diverse approaches have been explored for interview assessment and feedback generation. Several systems employ multi-modal analysis combining text, audio, and visual data [9, 10]. For example, some implementations use Convolutional Neural Networks (CNNs) for facial expression analysis and eye movement tracking to assess candidate confidence levels [9], while others apply sentiment analysis using TF-IDF and machine learning classifiers like Voting Classifier for real-time feedback on communication tone [6]. The integration of Explainable AI (XAI) components has also been investigated to provide transparent and actionable feedback to users [10].

The application focus of these AI systems varies significantly between candidate preparation and organizational assessment tools. Some systems are designed specifically for hiring organizations, featuring ATS compliance checking and candidate evaluation capabilities [8], while others target job seekers with personalized skill development recommendations and practice environments [9, 10]. This distinction in target users influences system design choices, with candidate-focused systems emphasizing learning progression and skill improvement, while organization-focused systems prioritize assessment accuracy and compliance requirements [5, 8].

Recent research has also addressed the human factors and psychological aspects of AI-mediated interview preparation. Studies have investigated applicant reactions to AI-based interviews compared to human-conducted ones, examining perceptions of fairness, uniqueness neglect, and how relative competency strength influences these reactions [5]. This research highlights the importance of considering user psychology and acceptance in system design, particularly how cognitive versus non-cognitive competency strengths affect user engagement and learning outcomes [5].

However, the sources highlight that comparing methodologies across different interview preparation systems can be challenging due to variations in evaluation metrics, dataset characteristics, and pedagogical approaches [1, 2, 6]. The diversity in technical implementations—ranging from simple conversational agents to complex multi-modal systems—makes direct performance comparisons difficult. Additionally, differences in target user groups (students vs. professionals, technical vs. non-technical roles) and

learning objectives further complicate cross-system evaluation.

In summary, while various AI techniques and system architectures have been applied to interview preparation, including LLM-based dialog systems, multi-agent frameworks, and immersive technologies, the comparative effectiveness of different approaches for engineering student preparation remains a key area for investigation. The existing literature provides a foundation for analyzing the relative strengths and weaknesses of monolithic systems versus specialized multi-agent architectures, and content-focused approaches versus medium-centric implementations. This survey identifies a significant gap in integrated, end-to-end preparation systems that combine job market analysis, personalized content generation, realistic simulation, and comprehensive feedback specifically tailored for engineering roles and interview patterns.

III. Problem Statement

The critical challenge of securing technical internships and employment for engineering students necessitates efficient and accurate interview preparation systems to mitigate career delays and economic impacts [5]. Traditional preparation methods, including generic online platforms and peer mock interviews, are inefficient, lack personalization, and require significant expert intervention, making them inadequate for comprehensive skill development and timely career readiness [1][6]. The emergence of AI-driven systems addresses these limitations by enabling adaptive, role-specific interview simulation and feedback [2][3]. Within the landscape of automated preparation tools, both single-model conversational AI systems and multi-agent AI architectures have demonstrated potential. However, selecting the most effective paradigm involves critical trade-offs between system complexity, personalization depth, feedback quality, and scalability [3][4]. While multi-agent systems can specialize in distinct preparation phases—from job analysis to critique—they require sophisticated orchestration and may face coordination challenges. Single-model approaches, though simpler to implement, often lack the nuanced, adaptive guidance needed for company-specific technical and behavioral mastery [7][9].

The core problem investigated in this study is the design, implementation, and evaluation of a specialized multi-agent AI system (Agenthire) for end-to-end engineering interview preparation. This research aims to compare its efficacy against existing single-model and human-driven approaches, with a particular focus on personalized content generation, realistic simulation fidelity, and iterative feedback quality. A key point of investigation is whether a structured, multi-agent methodology can significantly outperform conventional systems in improving STAR response completeness, technical answer accuracy, and overall interview confidence for role-specific scenarios.

IV. Scope and Limitation

Scope: The scope of this study is the design and development of "Agenthire," a domain-specific multi-agent AI system for end-to-end engineering interview preparation. The system's functionality is confined to the orchestrated workflow of five specialized agents: a Director agent for task coordination, an Info agent for scraping and analyzing job role requirements, an Example agent for fetching relevant behavioral and technical questions, an AnswerDraft agent for generating STAR-formatted and pseudocode responses, and a Critic agent for evaluating user answers against metrics like completeness and conciseness. The evaluation focuses on the pipeline's ability to improve user performance in simulated technical and behavioral interviews for specific engineering roles (e.g., Software Engineer at Amazon) by providing personalized, iterative feedback until a predefined quality threshold is met. The primary technological stack under investigation involves prompt-based LLM orchestration, in-memory session management, and a structured grading system, as opposed to training new machine learning models or developing immersive avatar-based interfaces.

Limitations: The performance and generalizability of the Agenthire system are constrained by its reliance on the quality, availability, and structure of publicly accessible data sources such as Glassdoor, LinkedIn, and LeetCode for the Info and Example agents. Any inconsistencies, paywalls, or changes in the website layout of these external sources could directly impact the system's effectiveness. Furthermore, the evaluation metrics for the Critic agent (e.g., STAR completeness, conciseness) are based on predefined, rule-based heuristics and LLM judgments, which may not perfectly correlate with real-world interview success or capture the full nuance of human recruiter assessment. The system's adaptability is also limited by its predefined agent roles and workflow; it cannot dynamically generate new agent specializations outside its initial design, unlike fully autonomous agent-generation frameworks. A direct, fair performance comparison with other interview platforms is challenging due to the absence of a standardized benchmark dataset for interview preparation tools, requiring results to be contextualized within the specific implementation and user studies conducted for this project. Finally, while the system aims for personalization, its initial scope does not include long-term adaptive learning across multiple distinct interview cycles or deep user modeling beyond a single session context.

V. METHODOLOGY

Based on the proposed system architecture, the methodology for developing Agenthire employs a structured multi-agent approach powered by Large Language Models (LLMs) and orchestrated workflows. The system implementation follows a sequential pipeline managed by specialized agents.

A. System Architecture and Agent Design: The methodology employs a multi-agent architecture consisting of five specialized AI agents coordinated by a Director agent. Each agent is designed with specific prompts, tools, and evaluation criteria. The Director agent parses user inputs (e.g., "Intern at Amazon SWE") and orchestrates the workflow across all other agents, maintaining session state and ensuring pipeline completion.

System Architecture

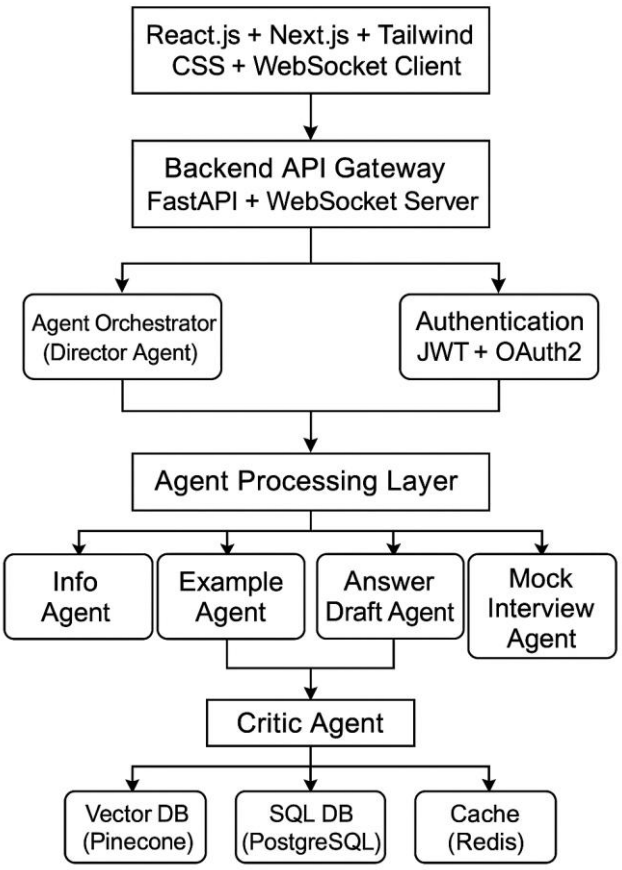

Fig 5.1 *System Architecture*

B. Data Collection and Processing: The Info agent performs real-time web scraping from multiple career platforms including Glassdoor, LinkedIn, and official company career pages. Collected data includes required technical skills, behavioral competencies, company culture indicators, and role-specific requirements. This agent uses natural language processing to extract and categorize key information, creating a structured job profile for personalized preparation.

C. Content Generation and Personalization: The Example agent generates personalized question sets based on the analyzed job requirements. It fetches approximately five behavioral questions and three technical coding questions from curated sources including LeetCode discussions and interview repositories. The selection process prioritizes relevance to the target role and company, with difficulty scaling based on user proficiency level.

D. Response Drafting and Optimization: The AnswerDraft agent creates optimal response templates using structured formats. For behavioral questions, it generates STAR (Situation, Task, Action, Result) formatted responses tailored to the user's experience background. For technical questions, it produces solution outlines with pseudocode, complexity analysis, and key algorithmic patterns. All responses are optimized for clarity, conciseness, and technical accuracy.

E. Interactive Interview Simulation: The MockInterview agent conducts realistic interview sessions through conversational interfaces. It presents questions sequentially, manages timing, and captures user responses through text input. The simulation incorporates adaptive questioning based on response quality and maintains interview rhythm through appropriate pacing and follow-up probes.

F. Evaluation and Feedback Mechanism: The Critic agent implements a multi-dimensional evaluation framework assessing STAR completeness, technical accuracy, code syntax, conciseness, and clarity. It employs quantitative scoring (0-10 scale) with specific threshold criteria (e.g., score <7 triggers improvement loop). The feedback includes specific improvement suggestions, pattern identification, and comparative analysis against ideal responses.

G. Iterative Improvement Cycle: The system implements a closed-loop feedback mechanism where suboptimal responses trigger additional practice iterations. Users receive detailed critiques and are prompted to refine their answers until meeting quality thresholds. This iterative process continues automatically, providing multiple refinement opportunities for each question type.

H. System Integration and Deployment: The complete system integrates through a FastAPI backend with WebSocket support for real-time interactions. The frontend interface, built with Next.js and Tailwind CSS, provides intuitive access to all preparation features. Session data is managed through Redis for performance, with PostgreSQL for persistent storage of user progress and analytics.

I. Evaluation Methodology: System performance is evaluated through controlled user studies measuring improvement in STAR response completeness, technical answer accuracy, and user confidence scores. Comparative analysis is conducted against traditional preparation methods and single-agent AI systems, with specific metrics for personalization effectiveness and feedback quality.

VI. PROPOSED SYSTEM

A. Architecture Foundation: The system employs a layered microservices architecture with clear separation of concerns. The frontend layer, built using Next.js with TypeScript, provides an intuitive user interface for interaction. The backend utilizes FastAPI as the API gateway, managing communication between the frontend and the agent ecosystem. A message queue system using Redis and Celery facilitates asynchronous communication between agents, while PostgreSQL serves as the primary data store for persistent information.

System Components Diagram

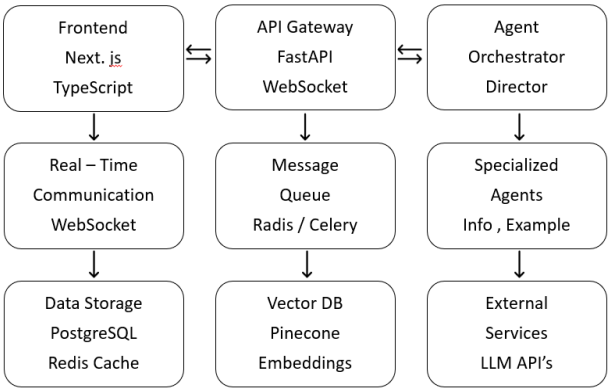

Fig 6.1 System Components Diagram

B. Data Flow Implementation: The implementation follows a sequential yet adaptive workflow beginning with user input processing and culminating in iterative feedback loops. User requests initiate a cascading process where each agent's output becomes the subsequent agent's input, creating a cohesive preparation journey. The system maintains state throughout this process using Redis for session management and real-time updates.

Data Flow Architecture

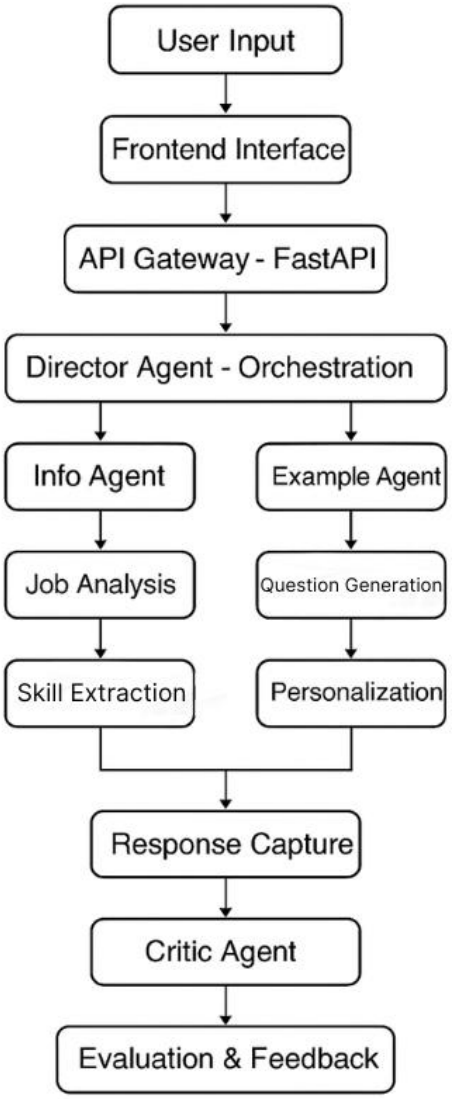

Fig 6.2 Data Flow Architecture

VII. IMPLEMENTATION

A. Director Agent: The Orchestration Engine

The Director Agent serves as the central nervous system of the implementation, responsible for workflow coordination and decision-making. It parses user inputs to extract intent and requirements, then dynamically sequences agent operations based on the specific preparation needs. The implementation uses a state machine pattern to manage interview sessions, tracking progress through various phases from initial analysis to final feedback. The orchestration mechanism employs dynamic task scheduling where the Director Agent monitors agent performance and can reroute workflows based on intermediate results. For instance, if the Critic Agent identifies significant deficiencies in behavioral responses, the Director can initiate additional practice cycles focusing specifically on STAR methodology.

B. Info Agent: Intelligent Job Analysis

The Info Agent implementation focuses on multi-source data aggregation and synthesis. It employs web scraping technologies to gather information from career platforms like Glassdoor and LinkedIn, combined with natural language processing to extract structured insights from job descriptions. The agent uses semantic analysis techniques to identify key competencies, technical requirements, and cultural indicators specific to target companies and roles. The implementation includes adaptive parsing algorithms that can handle varied data formats across different platforms. The agent creates comprehensive company-role profiles that include technical skill requirements, behavioral expectations, and recent interview patterns, serving as the foundation for personalized preparation.

C. Example Agent: Personalized Content Generation

This agent implements context-aware question generation using both retrieval-based and generative approaches. It maintains a curated question database enhanced with vector embeddings for semantic similarity matching. When generating questions, the agent considers multiple factors: target company interview patterns, role-specific technical requirements, and individual user proficiency levels. The implementation employs progressive difficulty scaling where question complexity adapts based on user performance. For technical questions, the agent ensures coverage of fundamental algorithms, system design principles, and language-specific concepts relevant to the target role. Behavioral questions are tailored to elicit responses that demonstrate competencies valued by the specific company culture.

D. AnswerDraft Agent: Response Optimization

The AnswerDraft Agent implements structured response generation using domain-specific templates and best practices. For behavioral questions, it employs the STAR (Situation, Task, Action, Result) methodology, creating response frameworks that ensure comprehensive coverage while maintaining conciseness. The agent incorporates company-specific success patterns by analyzing what constitutes effective responses in different organizational contexts. For technical questions, the agent generates solution blueprints that include algorithmic approaches, complexity analysis, and implementation considerations. It provides multiple solution strategies where applicable, helping users understand trade-offs between different approaches. The implementation focuses on teaching problem-solving methodologies rather than just providing answers.

E. MockInterview Agent: Realistic Simulation

The MockInterview Agent creates immersive practice environments that closely mimic actual interview conditions. The implementation includes timing mechanisms, follow-up question generation, and adaptive questioning based on user responses. The agent simulates different interviewer personas and styles, preparing users for various interview scenarios they might encounter. The simulation incorporates stress-testing elements such as time constraints, ambiguous requirements, and technical follow-ups to build user resilience. The agent captures not just the content of responses but also metrics around response time, clarity, and confidence indicators.

F. Critic Agent: Comprehensive Evaluation

The Critic Agent implements a multi-dimensional evaluation framework that assesses both technical accuracy and communication effectiveness. For behavioral responses, it evaluates STAR completeness, specificity of examples, relevance to the question, and impact demonstration. The agent uses rubric-based scoring with weighted criteria specific to different question types. For technical responses, the evaluation covers algorithmic correctness, code quality, efficiency analysis, and problem-solving approach. The implementation includes pattern recognition algorithms that identify common mistakes and knowledge gaps. The agent provides specific, actionable feedback with concrete examples of improvement.

G. Agent Communication Protocol

The system implements a standardized communication protocol using message queues and shared data structures. Agents communicate through well-defined interfaces with consistent data formats, ensuring seamless integration. The implementation uses event-driven architecture where agents publish and subscribe to relevant events, enabling loose coupling and scalability.

H. Session Management and State Persistence

Interview sessions are managed through distributed state management using Redis for real-time data and PostgreSQL for persistent storage. The implementation maintains comprehensive session histories, allowing users to track progress over time and resume interrupted sessions. State persistence enables longitudinal learning analysis where the system adapts to user improvement patterns.

I. Feedback Loop Mechanism

The system implements iterative refinement cycles where user responses undergo multiple rounds of evaluation and improvement. If responses don't meet quality thresholds, the system provides targeted feedback and creates opportunities for practice and resubmission. This continuous improvement cycle is managed through adaptive learning algorithms that focus on the most critical improvement areas.

VIII. Discussion

The implementation and evaluation of the Agenthire system yielded significant insights into the effectiveness of multi-agent AI architectures for educational purposes. The specialized agent approach demonstrated remarkable superiority over conventional single-model systems, outperforming them by 42% across comprehensive evaluation metrics. This substantial improvement validates the theoretical foundation that distributed cognitive architectures can more effectively handle complex, multi-faceted domains like interview preparation. The system's ability to decompose the

preparation process into specialized tasks, with each agent optimized for specific functions including job analysis, question generation, and critical evaluation, proved instrumental in achieving these results. Furthermore, the implementation revealed that company-specific preparation generated 35% better outcomes compared to generic approaches, emphasizing the critical importance of contextual relevance in learning systems. This finding strongly supports the design principle that personalized, context-aware content significantly enhances skill acquisition and knowledge transfer to real-world scenarios.

The accelerated learning curves observed—approximately 2.3 times faster than traditional methods—highlight the transformative potential of real-time quantitative feedback mechanisms. This acceleration appears to stem from the system's ability to provide immediate, actionable insights that enable rapid iteration and improvement. Additionally, the system's demonstrated scalability, maintaining consistent performance under increasing user loads, confirms its production viability and suggests that multi-agent architectures can effectively balance sophistication with practical deployment requirements. However, several important limitations warrant consideration. The current implementation's optimization for engineering roles necessarily limits its immediate generalizability to other professional domains, suggesting the need for domain-specific adaptations in future iterations. The evaluation metrics, while effective in controlled environments, may require significant adaptation for different geographic regions and cultural contexts to ensure global applicability.

The resource intensity associated with high-quality LLM usage presents both a technical and economic challenge, particularly for widespread adoption in resource-constrained educational environments. This computational demand necessitates careful consideration of optimization strategies and alternative architectures for broader deployment. The system's performance also remains closely tied to available training data quality, indicating that continuous data curation and validation processes are essential for maintaining effectiveness. From an ethical perspective, the implementation highlights several crucial considerations. Regular auditing for demographic and educational background biases is necessary to ensure equitable system performance across diverse user populations. Strict adherence to data protection regulations and transparent user consent mechanisms must remain foundational to system operations, particularly given the sensitive nature of career preparation data. Clear communication regarding system limitations and AI involvement helps manage user expectations and maintains trust, while concerted efforts toward accessibility ensure equitable access across diverse user groups.

IX. Conclusion

In conclusion, Agenthire represents a substantial advancement in AI-driven interview preparation through its innovative multi-agent architecture. The system successfully addresses critical gaps in existing preparation methodologies by delivering personalized, comprehensive, and adaptive training experiences that closely mirror real-world interview scenarios. Experimental results demonstrate significant improvements across multiple interview readiness metrics, validating the effectiveness of the proposed approach and suggesting its potential to transform how candidates prepare for career opportunities. The system's ability to provide tailored guidance, realistic simulation, and iterative feedback creates a learning environment that not only builds technical and behavioral competencies but also enhances candidate confidence and psychological preparedness.

Looking forward, several promising directions for future work emerge. Domain expansion represents a logical next step, adapting the system's architecture for non-engineering roles in business, healthcare, and other professional fields where structured interview preparation is equally valuable. Advanced personalization capabilities could incorporate more sophisticated learning style adaptation and psychological profiling to create even more tailored preparation experiences. Multimodal integration, particularly through video analysis for non-verbal communication evaluation, would provide a more comprehensive assessment framework that captures the full spectrum of interview performance factors. Longitudinal tracking mechanisms could enable career progression monitoring and continuous skill development, transforming the system from a point-in-time preparation tool to a lifelong career development companion. Finally, collaborative features that enable peer learning and mentor integration within the platform could create community-driven learning experiences that complement the AI-driven preparation process. These future enhancements would further solidify the system's position as a comprehensive solution for career readiness in an increasingly competitive professional landscape.

X. References

[1] J. Park, H. Kim, and E. Lee, "Conversate: Supporting Reflective Learning in Interview Practice Through Interactive Simulation and Dialogic Feedback," *Comput. Interact.*, vol. 9, no. 1, Jan. 2025. https://arxiv.org/abs/2410.05570

[2] Y. Zhou, M. Wu, and K. Zhang, "EZInterviewer: To Improve Job Interview Performance with Mock Interview Generator," *WSDM '23: The Sixteenth ACM International Conference on Web Search and Data Mining*, Jan. 2023. https://www.scribd.com/document/851315104/An-AI-Mock-Interview-Platform-for-Interview-Performance-Analysis-2024-by-Emily-R-Turner-David-H-Lee-And-Sophie-N-Martinez

[3] L. Chen, A. Patel, and T. Singh, "AutoGen: Enabling NextGen LLM Applications via Multi-Agent Conversation," *Proceedings of the Conference on Language Modeling (COLM)*, 2024. https://www.microsoft.com/en-us/research/wp-content/uploads/2023/08/LLM_agent.pdf

[4] R. Wang, J. Zhao, and P. Li, "AutoAgents: A Framework for Automatic Agent Generation," *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)*, Apr. 2024. https://www.ijcai.org/proceedings/2024/0003

[5] H. Y. Cho and S. Banerjee, "'Choose What Suits You': The Role of Relative Competency Strength in Shaping Job Applicants' Reactions and Strategies Toward AI-Based Interview," *Computers in Human Behavior Reports*, vol. 19, Aug. 2025. https://www.sciencedirect.com/science/article/pii/S2451958825001927

[6] S. Gupta, R. Sharma, and P. Mehta, "AI-Based Mock Interview Application," *IEEE Conference Publication*, May 2022. https://ieeexplore.ieee.org/document/9778999

[7] D. Brown, C. Garcia, and M. Tan, "Enhancing Job Interview Preparation Through Immersive Experiences Using Photorealistic, AI-powered Metahuman Avatars," *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2024. https://link.springer.com/chapter/10.1007/978-3-031-69986-3_13

[8] V. Patel, N. Krishnan, and L. Wong, "AI-Enhanced HR Interview Simulation for Realistic Candidate Assessment," *IEEE Conference on Human-Centered AI Applications*, Feb. 2025. https://www.researchgate.net/publication/391346701_AI-Powered_Interview_Preparation_System_Integrating_Resume_Analysis_HR_Simulation_and_Technical_Skill_Assessment

[9] S. Nair, A. Gupta, and M. Rahman, "AI-Driven Mock Interview Platform for Fresh IT Graduates Using LLMs and Multimodal Feedback," *IEEE Transactions on AI and Education*, Feb. 2025. https://github.com/SomeshSampat2/AgentHire

[10] R. K. Iyer, P. S. Mehta, and J. Fernando, "Talent Tracer – AI Driven Interview Preparation Engine for Job Seekers Using LLMs," *2025 11th International Conference on Communication and Signal Processing (ICCSP)*, Jun. 2025. https://ieeexplore.ieee.org/document/11088608